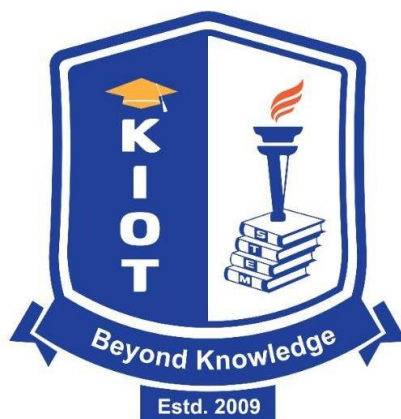


KNOWLEDGE INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

(Approved by UGC & AICTE, New Delhi and Affiliated to Anna University, Chennai)
Kakapalayam (PO), Salem – 637 504.



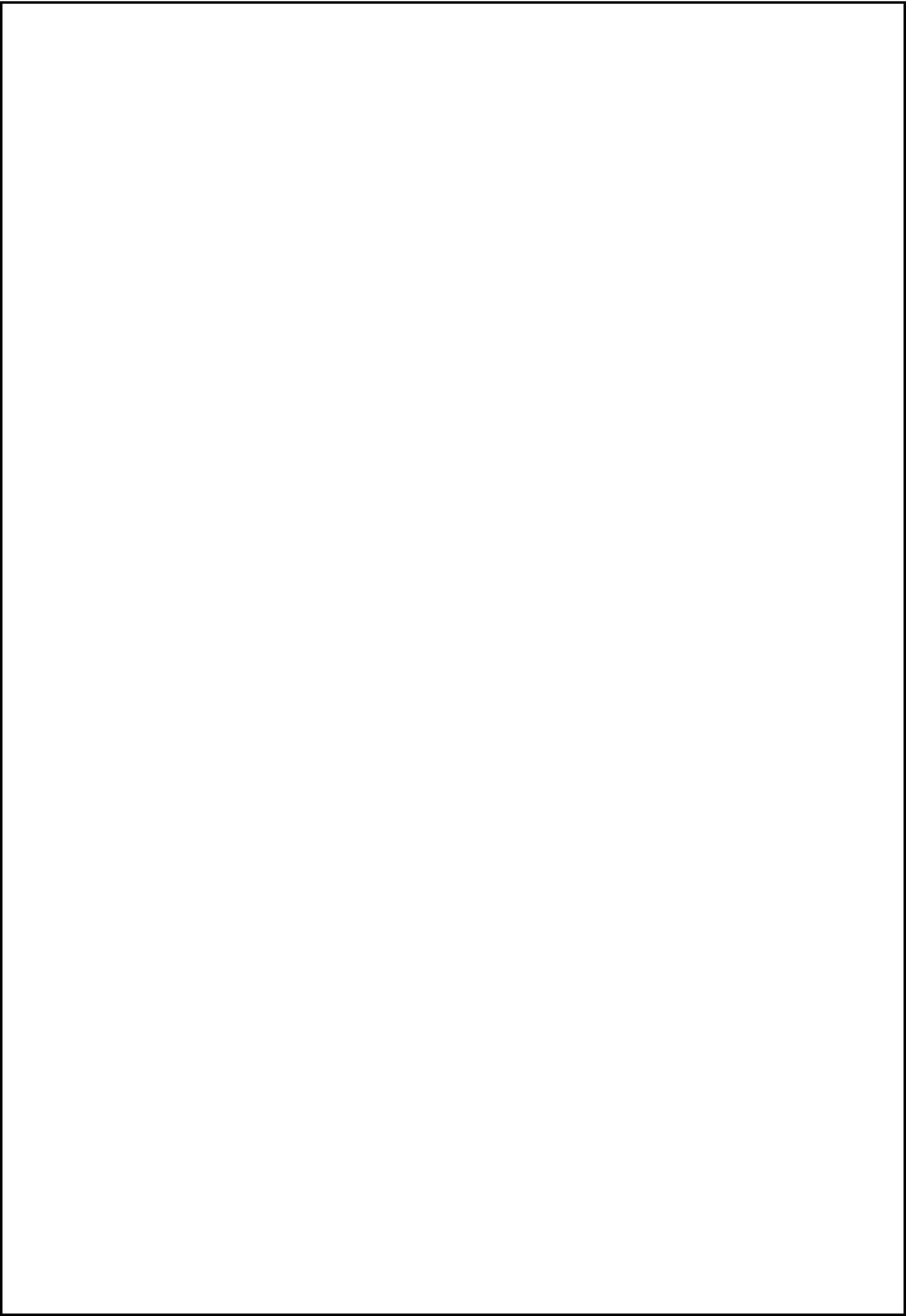
DEPARTMENT OF MECHANICAL ENGINEERING

INTERGRATED PRODUCT DEVELOPMENT

INSTITUTE TO INDUSTRY (I2I)

I2I TRAINING PROGRAM ABSTRACT

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Name of the resource person:	Mr. Shanmugam S
Module:	IV
Title:	Model Based System Engineering
Training Period:	27/01/2025 – 01/02/2025
Year / Sem / Sec / Regulation	III / V / B / 2023
Academic Year	2025



Vision and Mission of the Institute

Vision of the Institute

To be a world class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities.

Mission of the Institute

A	Promote academic growth by offering state-of-art undergraduate, postgraduate and doctoral programs and to generate new knowledge by engaging in cutting – edge research
B	Nurture talent, entrepreneurship, all-round personality and value system among the students and to foster competitiveness among students
C	Undertake collaborative projects which offer opportunities for long-term interaction with academia and industry
D	Pursue global standards of excellence in all our endeavors namely teaching, research, consultancy, continuing education and support functions

Vision and Mission of the Department

Vision

To create competent and industry relevant Mechanical Engineers with professional and social values to meet global challenges.

Mission

M1	Enabling environment for effective teaching - learning and research to meet global challenges.
M2	Motivating students to pursue higher education and to excel in competitive examinations and entrepreneurship.
M3	Establish a continuous Industry Institute Interaction to make the students employable.
M4	Inculcate the students' leadership quality with ethical values and spirit of team work.

Programme Educational Objectives (PEOs)

PEO 1	Graduates will apply the knowledge of Mechanical Engineering to solve real world Engineering problems.
PEO 2	Graduates will have the required attributes to pursue advanced education in Engineering and Technology.
PEO 3	Graduates will have the leadership skills with ethical values and team spirit.

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

Program Specific Outcomes (PSOs)	
After the successful completion of B.E. Programme in Mechanical Engineering, the graduates will able to	
PSO 1	Apply the knowledge of Computer Aided Design and Computer Aided Engineering tools to design and analyze the products and process related to Mechanical Engineering systems.
PSO 2	Develop the knowledge and skill relevant to Heating Ventilation and Air-Conditioning industries.
PSO 3	Exhibit the ability to make a product related to Mechanical Engineering and allied engineering fields.

LIST OF CONTENTS / EVENTS OF THE TRAINING PROGRAM

DAY 1	27/01/2025 (Monday)
FORE NOON	
❖ . An introduction to Systems Engineering was provided, highlighting its importance in managing complex projects effectively.	
❖ The session emphasized the value of concept maps, explaining that every course or project should begin with a clear understanding of key concepts.	
❖ . A structured, step-by-step learning approach was introduced: first grasping the concepts, then exploring their use, and finally applying them in practical projects.	
AFTER NOON	
❖ The importance of following proper procedures in project work was discussed — without the right steps, projects are more likely to fail.	
❖ The session reinforced that visual planning tools, such as concept maps, enhance idea connection and support better project planning.	
❖ The Project-Based Learning (PBL) method was introduced, which connects classroom learning with real-world applications to improve understanding and engagement.	
DAY 2	28/01/2025 (Tuesday)
FORE NOON	
❖ Explored how to excel in mathematical modeling, focusing on representing real-world systems using mathematical concepts.	
❖ Covered key topics such as units and dimensions, the concept of time derivatives, and dimensional analysis.	
❖ Gained an understanding of scaling techniques used in mathematical modeling to simplify complex problems.	
AFTER NOON	

❖ Understood the process of building a mathematical model, demonstrated through the Tower Problem as a practical example.	
❖ Got introduced to the fundamentals of Microsoft Excel, with a focus on its role in modeling and data analysis.	
❖ Applied this knowledge through basic Excel practice exercises to reinforce concepts and improve practical skills.	
DAY 3	29/01/2025 (Wednesday)
FORE NOON	
❖ Discussed systems of equations and the process of matrix multiplication as tools for solving mathematical models.	
❖ Learned how to create graphs using matrix values to visualize relationships and behaviors in systems.	
❖ Explored linear, quadratic, and cubic models commonly used in mathematical modeling to represent different types of system behavior.	
AFTER NOON	
❖ Studied the concept of linear stability analysis, including key ideas such as: ❖ Steady state ❖ Equilibrium solutions	
❖ Introduced to Lyapunov stability, covering: ❖ The quadratic form ❖ Lyapunov conditions for assessing local stability	
DAY 4	30/01/2025 (Thursday)
FORE NOON	
❖ Discussed the Suez Canal problem in detail.	
❖ Broke down the problem into key components.	
❖ Understood the underlying causes and implications of the problem	
AFTER NOON	
❖ Analyzed the problem to identify potential solutions.	
❖ Began preparing a PowerPoint presentation based on the identified solution.	
❖ Conducted background research to support and validate the proposed solution.	

DAY 5	31/01/2025 (Friday)
FORE NOON	
❖ Sir gave insights on how to effectively deliver a presentation.	
❖ Each team presented their PowerPoint (PPT) slides team-wise.	
❖ While one team was presenting, another team was assigned to evaluate the presentation.	
AFTER NOON	
❖ The team presentations continued during the afternoon session.	
❖ After each presentation, the evaluating team provided feedback based on content, communication, and design.	
❖ Suggestions were shared to help improve the quality of future presentations.	
DAY 6	01/02/2025 (Saturday)
FORE NOON	
❖ We learned how to build and understand a continuous model.	
❖ We studied the predator-prey model, which explains how animals like predators and their food (prey) interact.	
❖ We also learned about the two-species competition model, where two different species compete for the same resources.	
AFTER NOON	
❖ Understood how species interact using mathematical models.	
❖ Discussed the results of predator-prey and competition models.	
❖ Learned how to analyze stability in these models.	

I2I CO-ORDINATOR

HOD/MECH

PRINCIPAL

EXECUTIVE CHAIRMAN